

KESHAV KUMAR KARN

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Summary

Computer Science Engineering student focused on software engineering, with hands-on experience building native Android applications, full-stack web platforms, backend systems, and cloud-connected IoT applications. Strong foundation in C++, Data Structures & Algorithms, Object-Oriented Programming, and Database Management Systems, with additional experience in applied Machine Learning and Computer Vision.

Technical Skills

Languages: C++, C, Java, Kotlin, Python, JavaScript, TypeScript, SQL

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS)

Backend & Web: Node.js, Express.js, FastAPI, RESTful APIs, React.js, Next.js, Svelte/SvelteKit, PostgreSQL, MySQL, Tailwind CSS

AI/ML & Computer Vision: PyTorch, CLIP, Sentence-Transformers, XGBoost, Scikit-learn, Transformers, NLP, Computer Vision (OpenCV)

Cloud, IoT & Tools: AWS IoT Core, MQTT (TLS), Docker, pgvector, SQLAlchemy, Git/GitHub, Linux, Network Security & Threat Management

Education

Thapar Institute of Engineering & Technology

Patiala, Punjab, India

Bachelor of Technology in Computer Science and Engineering

Expected May 2027 **CGPA: 6.78 / 10.00**

Projects

GramFlow — Native Android Inventory Management App

[\[GitHub\]](#)

Kotlin, Jetpack Compose, Android SDK, Room DB, Coroutines, Material 3

- Engineered an offline-first native Android POS and inventory management application using Kotlin and Jetpack Compose, supporting real-time batch tracking and customer debt ledger bookkeeping.
- Designed an automated FIFO batch liquidation engine implementing per-gram cost tracking and gross profit calculation, backed by Room/SQLite persistence.
- Implemented atomic transaction rollback logic to preserve data integrity across concurrent inventory and ledger updates.

GramFlow — Web Terminal & FIFO Ledger Suite

[\[GitHub\]](#)

Next.js, React, TypeScript, PostgreSQL, Tailwind CSS

- Architected a full-stack web terminal with automated FIFO cost-of-goods calculation and double-entry audit logging for inventory transactions.
- Implemented role-based access control (RBAC) to enforce authorization boundaries across terminal operations.
- Optimized relational database schema constraints and query design in PostgreSQL to maintain ACID transaction consistency.

De-Insure — Parametric Insurance Framework for Cold Chain Logistics

[\[GitHub\]](#)

Python, PyTorch, XGBoost, ESP32, AWS IoT Core, MQTT, React

- Built an IoT telemetry pipeline using ESP32, DHT22, MPU6050, and NEO-6M GPS sensors, streaming transit metrics over MQTT (TLS) to AWS IoT Core.
- Trained an XGBoost classifier for spoilage detection and a PyTorch LSTM model for shelf-life estimation from live telemetry.
- Integrated cloud-ingested telemetry and model inference into a React dashboard for real-time cold-chain monitoring.

FINDORA — AI-Powered Multimodal Lost & Found Portal

[\[GitHub\]](#)

FastAPI, Svelte 5, PostgreSQL, pgvector, PyTorch, CLIP, Docker, AWS

- Built a full-stack multimodal AI portal for university campus item recovery, pairing lost and found reports using cross-modal retrieval across image-to-image, text-to-image, and text-to-text matching within a joint embedding space.
- Implemented a transformer embedding pipeline using clip-ViT-B-16 (512-dim multimodal) and all-MiniLM-L6-v2 (384-dim text), with PostgreSQL pgvector HNSW indexes for sub-millisecond cosine similarity retrieval.
- Engineered a secure contact escrow system using JWT-authenticated role-based access control (RBAC), async background match scoring with domain heuristics (campus proximity graphs, keyword extraction, color validation), and automated match archival.

Certifications

- Cyber Threat Management — Cisco Networking Academy, TIET
- Network Defense — Cisco Networking Academy, TIET
- Real-Time Data Analytics — Experiential Learning Centre (ELC), TIET
- Real-Time Application Based on Computer Vision — ELC, TIET
- Secure Network Configuration — ELC, TIET

Sep 2026

Feb 2026

2025

2025

2024